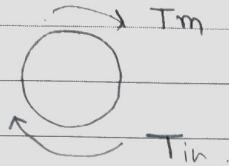


## DERIVATIONS

step.1 · Derive the model one motor

FBD



$$\Sigma T = J \frac{dw}{dt}$$

$$T_{\text{motor}} - T_{\text{frict}} - T_{\text{load}} = J \frac{dw}{dt}$$

↓  
Input torque

T = torque / motor / load

J = Inertia

$\frac{dw}{dt}$  = Change in  $w$

$w$  = angular.

$$T = \frac{1}{2} J w^2$$

Solving :  $T_f = Bw$  } Energy loss due to friction

Substitute in equation

$$J \frac{dw}{dt} = T_{\text{motor}} - Bw - T_{\text{load}}$$

$$P = Bw^2$$

## Step 2 Electrical Side

For a DC motor Voltage Equation :  $V = L \frac{di}{dt} + Ri + e_b$

~~Servo motor~~

Servo Motor

Stepper Motor

Voltage eqn of DC motor

$$V = L \frac{di}{dt} + R_i + E_b$$

Using Kirchhoff's Voltage Law  
Sum of voltage in loop = 0

$L$  = Inductance

$R_i$  = Resistance

$\frac{di}{dt}$  = Change in  
current  
flowing in coil

$E_b$  = Back Emf  
(generated voltage)  
for rotation

Back emf relation

$$E_b = K_e \omega$$
$$T_m = K_t i$$

} Faster rotation  $\rightarrow$  higher Emf

Combining Equation  $\Rightarrow V = L \frac{di}{dt} + R_i + K_e \omega$  ~~[KVL]~~

2nd order derivative equation

Step 3: Combining Electrical + Mechanical logic

mechanical  
Electrical  $\Rightarrow J \frac{d\omega}{dt} = K_t i - B\omega - T_{load}$

Electrical  $\Rightarrow V = L \frac{di}{dt} + R_i + K_e \omega$

Graph shows curves not straight lines

### Step 3.

Combining Electrical And Mechanical Equation

$$V(t) = L \frac{di(t)}{dt} + Ri(t) + Ke w(t) \quad - \text{electrical}$$

$$J \frac{dw(t)}{dt} = T_{\text{motor}} - Bw(t) - T_{\text{load}}(t) \quad - \text{mechanical}$$

depended - Current  $\rightarrow$  Torque  $\rightarrow$  Rotation - E.M.F

current  $\Rightarrow i(t)$

$$T_{\text{motor}} = K_t i(t)$$

So mechanical eqn:  $J \frac{dw}{dt} = K_t i - Bw - T_{\text{load}}$

$$\Rightarrow K_t i = \frac{Jdw}{dt} + Bw + T_{\text{load}}$$

Current is expressed  $\leftarrow i(t) = \frac{Jdw/dt + Bw + T_{\text{load}}}{K_t}$   
in terms of speed.

$$V = L \frac{di}{dt} + Ri + Ke w$$

Substituting  $i = \frac{Jdw/dt + Bw + T_{\text{load}}}{K_t}$  in electric eqn

$$\Rightarrow V(t) = a_2 \frac{d^2 w}{dt^2} + a_1 \frac{dw}{dt} + a_0 w$$

$a_1 \Rightarrow$  depend on  $R, B, L$

$a_2 \Rightarrow$  depend on  $L, J$

$a_0 \Rightarrow$  depends on friction + back emf

So now Electrical + Mechanical equation are turned into single differential equation  $w(t)$  with one output variable (speed).  
↓  
2<sup>nd</sup> order

$\frac{w(t)}{V(t)} \Rightarrow$  Output (How speed responds to voltage)

$$\frac{w(s)}{V(s)} = \frac{K}{(Js + B)(Ls + R) + K^2}$$

} Transfer function for DC motor

SI

Step(4) Use Laplace Transform  
time domain  $\rightarrow$  s domain

$$\frac{d}{dt} \Rightarrow s \quad i(t) \Rightarrow I(s) \quad w(t) \Rightarrow \Omega(s)$$

$$\text{Electrical equation} \Rightarrow V(s) = L \frac{di}{dt} + R i + K_e \Omega$$

$$\text{after substitution } V(s) = L s I(s) + R I(s) + K_e \Omega(s)$$

Mechanical eqn after substitution

$$\Rightarrow J s \Omega(s) = K_t I(s) - B \Omega(s) \quad [\text{load removed for simpler eqn}]$$

From mechanical equation

$$\rightarrow J s \Omega(s) + B \Omega(s) = K_t I(s)$$

$$\rightarrow \text{Factor : } \Omega(s) (J s + B) = K_t I(s)$$

$$\text{Now : } I(s) = \frac{(J s + B) \Omega(s)}{K_t}$$

Substituting into Electrical.

$$V(s) = L s I(s) + R I(s) + K_e \Omega(s)$$

$$V(s) = L s \left( \frac{(J s + B) \Omega(s)}{K_t} \right) + R \left( \frac{(J s + B) \Omega(s)}{K_t} \right) + K_e \Omega(s)$$

$$V(s) = \Omega(s) \left[ \frac{(Ls+R)(Js+B) + K_e}{K_t} \right]$$

$$K_t V(s) = \Omega(s) [(Ls+R)(Js+B) + K_e K_t]$$

Final transfer function

$$\frac{\Omega(s)}{V(s)} = \frac{K_t}{(Ls+R)(Js+B) + K_e K_t}$$

$$= \frac{K_t}{LJs^2 + (LB+RJ)s + (RB+K_e K_t)}$$

$\swarrow$   
 2<sup>nd</sup> Order  
 (energy storage)

$\downarrow$   
 First Order  
 (energy loss)

$\searrow$   
 Constant  
 (steady-state)

Compare it with  $\Rightarrow \frac{K}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

## Step 5

$$\text{Error : } e(t) = \omega_{\text{ref}}(t) - \omega(t)$$



Desired Speed

Actual Speed.

PID Controller:

$$U(t) = K_p e(t) + K_i \int e(t) \cdot dt + K_d \frac{de(t)}{dt}$$



Reacts to current error.

Removes steady-state error.

Predicts future error and smoothers response

$u(t)$  = Voltage applied to motor

## Step - 6

\* Motor Transfer function      \* PID Laplace.

$$G(s) = \frac{w(s)}{V(s)} \rightarrow \text{Initial } s=0$$

for Laplace transform.

$$C(s) = K_p + \frac{K_i}{s} + K_d(s)$$

⇒ Closed loop Transfer Function.

$$\frac{w(s)}{w_{ref}(s)} = \frac{C(s) G(s)}{1 + C(s) G(s)}$$

$e(t)$  = sensor difference

PID output = PWM signal

$V(t)$  = voltage applied.

$w(t)$  = encoder feedback.

Energy Equation 